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International Journal of Performability Engineering

vol. 16, no. 8, August 2020, pp. 1235-1244
DOI: 10.23940/ijpe.20.08.p10.12351244

Analysis and Implementation of a QoS-Aware Routing Protocol based on a Heterogeneous Network

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Abstract

A heterogeneous network is an environment with different types of interconnected networks that contains a variety of communication links, networking modes, and communication protocols. The heterogeneity and variability of such networks have triggered a range of problems, such as security risks, unreasonable communication-resource allocation, the disordering of network access, and difficulties in ensuring quality-of-service (QoS) with business priorities. In this context, a QoS routing solution is proposed to combine the weighted-fair-queuing (WFQ) algorithm and an extended open-shortest-path-first (OSPF) protocol. The network model, multi-service transmission, and routing protocol of a marine-oriented heterogeneous network are simulated by means of OPNET. The proposed QoS routing solution is then verified in a high-load simulation environment. The experimental results prove that the solution can improve network resource control and routing optimization to some extent by reducing such QoS parameters as transmission delay of different services, average delay, and jitter of network.

Keywords: heterogeneous network (HetNet); QoS routing; OSPF; WFQ

(Submitted on February 25, 2020; Revised on May 16, 2020; Accepted on July 15, 2020)

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1. Introduction

In the wake of booming mobile-communication technology, it is unlikely that we can meet users' communication demands with a single mode or network of communication. In addition to the expansion of communication coverage, users are also requesting an improvement in the convenience and diversity of communication modes. In this circumstance, a heterogeneous network that integrates different types of communication modes and networks has become a research focus [1]. A heterogeneous network contains different types of inter-coordinated communication subnets, which differ from each other in their networking mode, network structure, technical parameters, control and management mechanism, and type of service (ToS). In addition, the subnets may belong to different owners (e.g., operators, enterprise private networks, and personal area networks), thus resulting in their heterogeneity, independence, and mutability. Therefore, communication transmission in a heterogeneous network does not involve simply solving the problems of a homogeneous network, but it must also cope with a range of emerging problems such as security risks, unreasonable communication-resource allocation, disordering of network access, and difficulties in ensuring QoS of priority service [2-4].

Generally, QoS is selected to evaluate the grade of service (GoS) that any network and computer system provides to specific applications and network flow. As an agreement regarding information transmission quality between network users, it acts as a performance measure of network and computer systems [5-6]. Moreover, QoS is also a critical research object in communications, which are based on heterogeneous networks. Its major applications in heterogeneous networks include QoS mapping and QoS routing, the latter of which plays a significant role in QoS-related studies.

The marine-oriented heterogeneous network, which consists of a communication-access network (CAN) and a backbone network, is used for the effective planning and unified scheduling of various resources and services in maritime emergency communication. It is able to guarantee the real-time and reliable transmission of multiple communication services (e.g., video

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communication, control information, and mission commands) by integrating a variety of communication technologies (e.g., LTE, satellite communications, and optical fiber) and physical platforms (e.g., base stations, UAVs, and ship-borne systems). Since the marine-oriented heterogeneous network uses highly maneuverable sites to form a dynamic topology, integrates multiple communication modes, and has complex communication links, it is necessary to satisfy such QoS constraints as time delay, jitter, and packet-loss ratio before any valid communication services are provided. Furthermore, the network has to work under high-traffic conditions when providing different types of services, so it is susceptible to traffic congestion and some other defects like large service-transmission delay and failure in real-time transmission. In this context, it is critical to ensure reasonable communication-resource planning, coordinated routing, and an orderly and efficient traffic flow by improving the QoS of heterogeneous networks.

2. QoS Demand Analysis for the Marine-Oriented Heterogeneous Network

As a CAN that connects "sky, space, coast and ocean," the marine-oriented heterogeneous network provides for the seamless, real-time transmission of different services (e.g., audio, video, command-and-control, and navigation information) by integrating a range of communication modes (e.g., satellite, microwave, and optical fiber) and communication platforms. As shown in Figure 1, its network topology consists of an IP network, a universal mobile telecommunications system (UMTS) and a wireless local area network (WLAN). The IP network serves as its core network and contains a network center and two bases. The network center is connected to the bases via optical-fiber links. A gateway router in the network center is connected to the Internet and UMTS. Ground stations of the bases are connected to wireless terminals (e.g., UAVs and ship-borne systems) to form a WLAN by way of microwave and satellite-communication links.

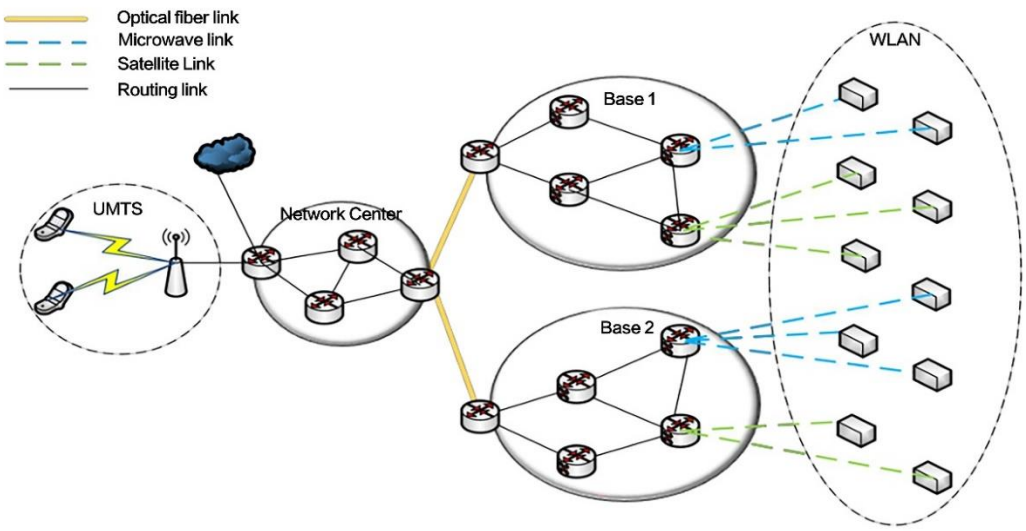

Figure 1. Topology of marine-oriented heterogeneous network

The marine-oriented heterogeneous network is designed to provide customized communication links, to ensure effective planning and scheduling of various resources and service demands in maritime communication, to develop user-centered resource service and management, and finally, to optimize maritime networking and communication services. Since it consists of a backbone network and a CAN and is capable of switching the service provider's network and exterior service network, an end-to-end QoS should be guaranteed for users. To this end, in view of HetNet's mobility, wireless transmission, and multihop routing, the following requirements should be satisfied to ensure the end-to-end QoS for users:

(1) Given that the network integrates multiple communication modes and has complex communication links, it is necessary to satisfy QoS constraints such as time delay, jitter, and packet-loss ratio before any valid communication services are provided.

(2) The network has to deal with different types of services including voice and video transmission. It must also meet varying performance requirements (e.g., bandwidths and time delay) during transmission. In addition, it is necessary to ensure the end-to-end transmission of important services in the first place according to their priorities.

(3) The network has some problems caused by high traffic and maldistribution of services: e.g., sustained high-traffic conditions of satellite communication systems, susceptibility to traffic congestion, and a large time delay in service

transmission. Therefore, a proper traffic-control mechanism should be adopted to ensure real-time service transmission.

(4) In marine-oriented HetNet, given that two users may select different communication modes across different types of networks, the network nodes need to dynamically coordinate available resources in communication networks and then select an appropriate communication path.

In HetNet, a variety of technical means and communication protocols should be utilized to transfer and translate the QoS demand among different types of networks. Moreover, the QoS demand may have different meanings and interpretations in different types of networks. A proper control mechanism should be adopted to manage the QoS demand when transmitted between different network layers. Furthermore, since the network layers play different roles in transmission, different QoS solutions should be used to meet diverse QoS demands [7-8].

A structural model of a hierarchical QoS (HQoS) is given in Figure 2. A conventional network structure consists of a physical layer, a data-link layer, a network layer, a transmission layer, and an application layer. Each network layer provides different services during communication, and the corresponding QoS also varies with layers. The QoS information of all layers is collected by the model to obtain network-state information, and the layers interact with each other by sharing the collected network-state information [9-10]. Once the network receives a QoS demand, the QoS guarantee is achieved through that each layer provides effective and reliable services. For example, services like voice, video, and data are monitored and administrated at the application layer; macro-control, including rate control and congestion control, is performed at the transmission layer based on the current network state; routing and resource management are implemented at the network layer by using reasonable QoS routing protocols; the data-link layer supports the QoS demand by enhancing backoff mechanisms and implements channel-assignment and admission control based on the current network state; and the physical layer meets relevant QoS parameter requirements via power control and speed matching. Given all of this, the present study intends to provide a QoS routing solution that satisfies QoS demands by following the routing protocol of a heterogeneous network in the network layer of a network model so that communication-resource control and routing optimization can be achieved via path selection and traffic control in a heterogeneous network.

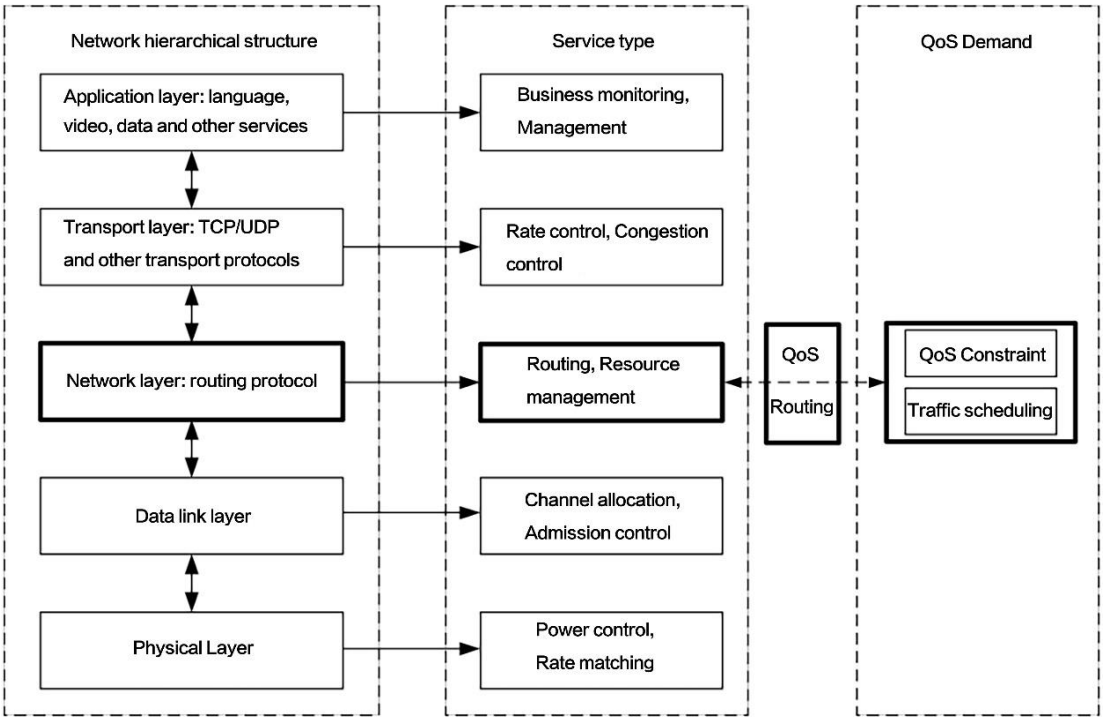

Figure 2. A structural model of HQoS

3. Design of QoS Routing Solution

Given that the subnets of a heterogeneous network all have their own characteristics, with complicated networking modes and diversified access technologies, different heterogeneous networks should select diverse QoS solutions for network performance optimization and reasonable resource allocation. Accordingly, a QoS routing solution that conforms to

characteristics of the marine-oriented HetNet should be applied to provide an end-to-end QoS guarantee for network communication. Since the marine-oriented HetNet is the integration of an IP network, a UMTS and WLAN, it has to deal with problems such as network resource maldistribution, high-traffic load, and difficulties in ensuring the reliable, end-to-end transmission of critical services. Therefore, we propose a QoS routing solution that introduces the weighted-fair-queuing (WFQ) algorithm of an IP network into the HetNet. In this way, network messages can be forwarded according to a particular scheduling policy by integrating with an OSPF extension protocol which is based on cost regulation and control. Also, an effective mechanism for traffic scheduling and routing forwarding will be applied to achieve network-resource control and routing-selection optimization.

3.1. Principle of WFQ

The WFQ algorithm is utilized to perform scheduling on the basis of a time stamp. Specifically, a time stamp which represents the time of dequeuing is assigned to packets when they enter a waiting queue during communication. As the scheduling task starts, the packet with the minimum time stamp is selected from the waiting queue, forwarded by a scheduler, and finally dequeued. The principle of WFQ is illustrated in Figure 3 [11-12].

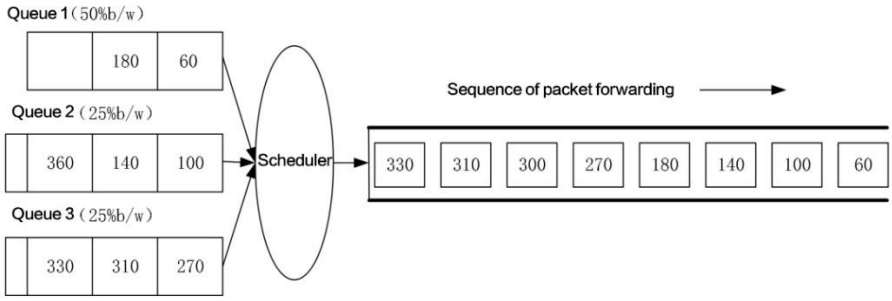

Figure 3. The principle of WFQ

The arrival or dequeuing of a packet is defined as an event. As defined, t_j stands for the occurrence time of the jth event and B_j represents relevant links. It is assumed that the operating rate of the scheduler is γ and that it runs continuously at this rate. The predicted bandwidth for completing the session j is denoted by ϕ_j, which can also be referred to as the weight. The virtual time is represented by $V(t)$, and assuming $V(0) = 0$.

$$V(t_{j-1} + \tau) = V(t_{j-1}) + \frac{\tau}{\sum_{j \in B_j} \phi_i}, \text{ where } 0 < \tau <= t_j - t_{j-1}, j = 2,3,4 \tag{1}$$

Each packet arriving at the scheduler is labeled with a virtual start-time stamp and a virtual end-time stamp. Accordingly, S_i^k and F_i^k are defined as the virtual start time and the virtual end time, respectively, of the kth packet of session i. P_i^k is defined as the kth packet of session i, L_i^k is defined as the length of the kth packet of session i, a_i^k is defined as the arrival time of the kth packet of session i, and γ_i is defined as the service rate of session i. In this case, S_i^k and F_i^k can be expressed as follows:

$$\begin{cases} S_i^k = Max\{F_i^{k-1}, V(a_i^k)\} \\ F_i^k = S_i^k + \dfrac{L_i^k}{\gamma_i} \end{cases} \tag{2}$$

Both the virtual start time and the virtual end time of packets arriving at the scheduler can be computed via the above equation. The packet with the minimum virtual end time is screened out by the scheduler and forwarded. Subsequent to each scheduling operation, B_j, a combination of the links should be maintained. In this case, the WFQ algorithm updates the virtual time function to ensure that B_j can be effectively maintained. If no packet arrives at the scheduler after time t and the arrival time $Next(t)$ of the next packet is a function of t—or in other words, if no packets arrive within the time range of (t, $Next(t)$)—then B_j should be maintained by updating the virtual time. Furthermore, $F_{\min}$ is defined as the least virtual completion time of a packet at time t. Based on Equation (2),

$$F_{min} = V(t) + \frac{Next(t) - t}{\sum_{j \in B_j} \phi_i} \tag{3}$$

Also,

$$Next(t) = t + [F_{min} - V(t)] \sum_{j \in B_j} \phi_i \tag{4}$$

Once an update operation is performed according to the virtual time in Equation (4) and the data packet is labeled with the virtual end time stamp, the scheduler will select a packet with the least virtual end time for forwarding and transfer [13-14].

3.2. OSPF Protocol Extension

The OSPF protocol conducts data transmission by using the computed shortest route while ignoring the resource utilization of the links with the shortest route (e.g., bandwidth). If there exists only one shortest route in the network, the same link or the same interface of a router may be used to perform packet forwarding during mass data transmission, thereby resulting in traffic congestion on this path. In brief, a conventional OSPF protocol exploits idle link resources rationally, which may result in traffic congestion on the shortest path in a real-life scenario, thus making it impossible to guarantee the QoS of the network.

The SPF algorithm suggests that path cost is critical to generating the shortest path. Routing calculation is associated with the cost of the router interface. The interface cost is configured according to the bandwidth of a link connecting the interface. Moreover, the bandwidth is inversely proportional to the cost. Therefore, based on the cost of the SPF algorithm, the link cost is adjusted in line with the link status to the extent that the algorithm can preferably adapt to network congestion [10]. The major adjustments are as follows: To comprehensively consider the link status of a communication network—especially the service conditions of link bandwidth which connect router interfaces—the interface cost is adjusted according to the bandwidth-availability ratio of the router interface. In this way, the shortest path based on interface cost can better suit the running status of the network and avoid link congestion effectively. In short, by regularly detecting the bandwidth-availability ratio of each interface through the router, the interface cost can be adjusted according to the running status of the interface so as to optimize the calculation of the shortest path and identify the optimal router that accords with the actual situation. The procedures of OSPF-based cost adjustment are shown in Figure 4.

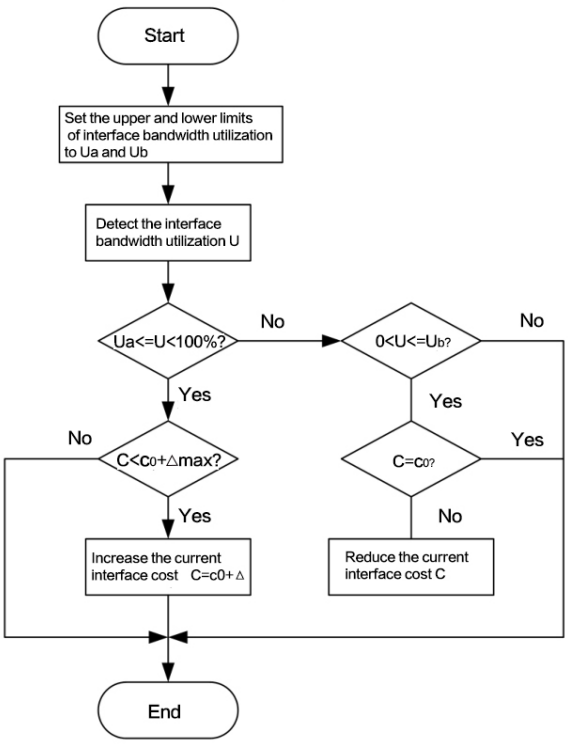

Figure 4. A flow chart of OSPF-based cost adjustment

Suppose that the bandwidth availability ratio of the router interface is $U(0 < U < 100\%)$ and set two thresholds: namely, the upper limit U_a and the lower limit U_b. The interface has three states based on the value of U: namely, a high-load state, an equilibrium state, and a low-load state.

(1) High-load state ($U_a \leq U \leq 100\%$): The cost of interface is low in the high-load state, so it is necessary to increase the cost. The interface cost can be expressed as follows,

$$C = c_0 + \Delta \tag{5}$$

Where, c_0 refers to the initial cost of interface, while Δ is the least increment of interface cost.

Moreover, the router releases router LSA in the area to synchronize with LSD. Since the router recalculates routing based on LSD and updates the corresponding routing list at set intervals, the optimal path computed by the SPF algorithm will change after the routing list is updated. In this way, busy network nodes can be bypassed to transfer network load [10].

(2) Equilibrium state ($U_b \leq U \leq U_a$): The cost of the interface in the equilibrium state shall not be adjusted. If adjusted (either increased or decreased), unnecessary network resources will be consumed to reduce network performance. Specifically, the service data will be constantly transferred, and the transmission path of services will also keep changing after adjustment. As a result, the routing list will be updated ceaselessly, and the router LSA will be frequently generated and released, thereby leading to excessive consumption of network resources. For these reasons, the interface cost shall remain unchanged in the equilibrium state.

(3) Low-load state ($0 \leq U \leq U_b$): The cost of interface in the low-load state can be processed in two ways. If the current interface cost is greater than the initial cost, it should be lowered. Since the current interface cost is above its initial value, it indicates that the interface cost was increased, or in other words, that the interface was overused at some point. Since the current bandwidth usability ratio of the interface is relatively low, the interface is able to provide more transmission services. Therefore, it is necessary to lower the current cost so that the links which connect the interface can be used effectively. However, if the current interface cost is equal to its initial value, no adjustment shall be made [10].

The detection period of the bandwidth-availability ratio of the interface should be set as the refresh time of router LSD to ensure that recalculated optimal routing can take effect immediately. In addition, the maximum increment of interface cost should be configured when the network is in the high-load state. The cost can be increased only when the total increment does not exceed its maximum value [11].

4. Experimental Simulation

In this section, network simulation is designed and implemented based on the OPNET Modeler.

As shown in Figure 1, the marine-oriented heterogeneous network consists of an IP network, a UMTS and WLAN. To be specific, the IP network, as the core network, is formed by a network center and two bases. The network center is connected to the bases via optical-fiber links. A gateway router in the network center is connected to the Internet and UMTS. Ground stations of the bases are connected to wireless terminals (e.g., UAVs and ship-borne systems) by virtue of microwave and satellite-communication links to form a WLAN. The network layer model is given in Figure 5.

The services used in simulation include email, file-transfer protocol (FTP), and voice. Table 1 lists the preset values of the three services.

As the heterogeneous network is characterized by different types of services and a large transmission flow, the service parameters are set as those in the high-load state during simulation.

Statistics are collected after a simulation model is set up, and all parameters are configured in OPNET. Different statistics reveal the performance parameters of different networks. Upon the completion of a simulation, the statistics are analyzed to determine whether the simulation satisfies related demands. Based on the service types and characteristics of the QoS parameters, the statistics are listed and described in Table 2.

The statistical charts of average transmission delay of email, FTP and voice are given in Figures 6, 7, and 8. The blue, green, and red curves stand for the average transmission delay in three scenarios: namely, the solution based on OSPF

(Only_OSPF), the solution based on WFQ (WFQ), and the QoS-routing solution based on WFQ and extended OSPF (QoS_Route). According to the results presented in Figures 6, 7, and 8, a statistical chart of average time delay is obtained (see Table 3).

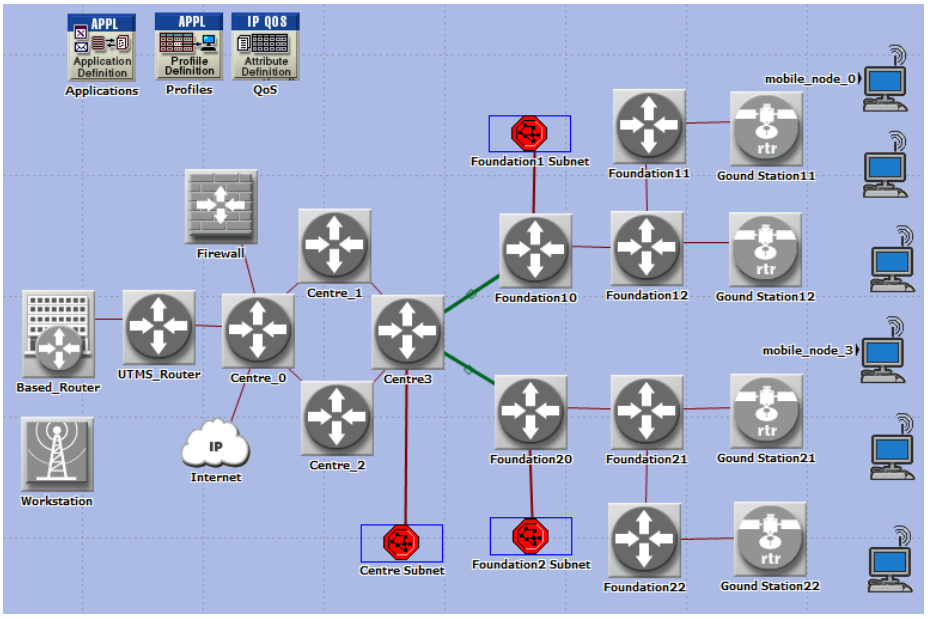

Figure 5. A network layer model of the marine-oriented heterogeneous network

Table 1. Preset values of services

Service Type	E-mail	FTP	Voice
Preset Value	● Low Load ● Medium Load ● High Load	● Low Load ● Medium Load ● High Load	● PCM Quality Speech ● Low- Quality Speech ● IP Telephone

Table 2. Simulation statistics and their description

Service Type	Statistics Name	Description
E-mail	Download Response Time(sec)	Includes the delay of connection establishment, transmission, and connection closure
FTP	Download Response Time(sec)	Includes the delay of connection establishment, transmission, and connection closure
Voice	Packet End-to-End Delay(sec)	Includes the delay of network encoding/decoding and compression/decompression.

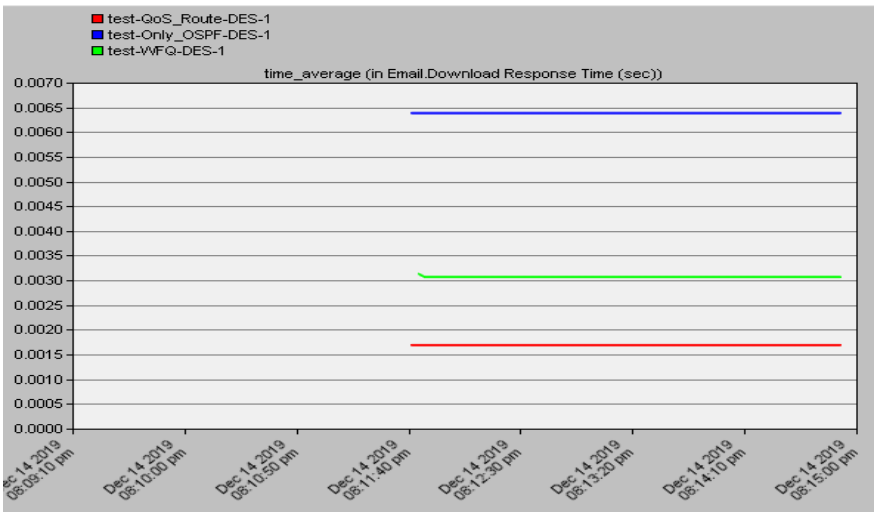

Figure 6. Comparison of average transmission delay of E-mail

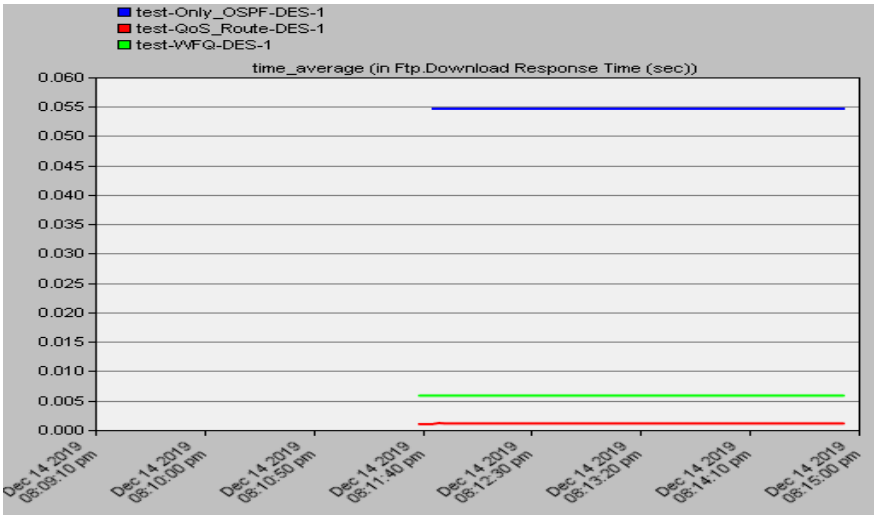

Figure 7. Comparison of average transmission delay of FTP

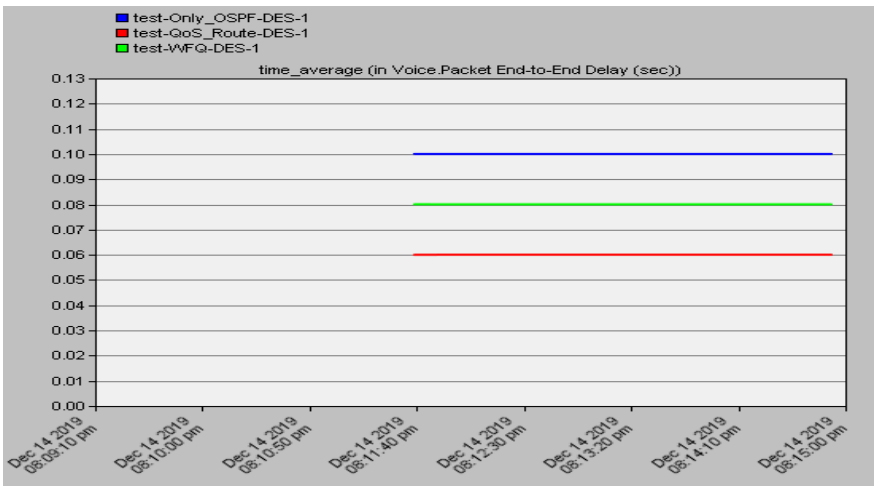

Figure 8. Comparison of average transmission delay of voice

Table 3. A statistical chart for average time delay of services at different scenarios

	Only_OSPF	WFQ	QoS_Route
E-mail	0.0065s	0.0030s	0.0017s
FTP	0.055s	0.006s	0.002s
Voice	0.10s	0.08s	0.06s

As can be seen from Table 3, the average transmission delay in the scenario with the QoS routing solution is significantly smaller than that of the other two scenarios. This finding indicates that the proposed QoS routing solution can better suit the heterogeneous network environment, meet the high QoS requirements of services, and improve resource utilization of the network to a certain degree.

Figures 9 and 10 show the average transmission delay and average transmission jitter of the heterogeneous network. The blue, red, and green curves stand for the statistical results in three scenarios: namely, the solution based on OSPF (Only_OSPF), the solution based on WFQ (WFQ), and the QoS routing solution based on WFQ and extended OSPF (QoS_Route).

The average time delay of services (see Table 4) was obtained by using the statistics of figures 9 and 10. As shown in Table 4, the time delay and jitter in the scenario with the QoS routing solution are obviously superior to those of the other two scenarios. This finding proves that the proposed QoS routing solution can better suit a heterogeneous network environment and can be used to control network resources, optimize routing, reduce network time delay and jitter, and improve QoS in high-load and high-traffic conditions.

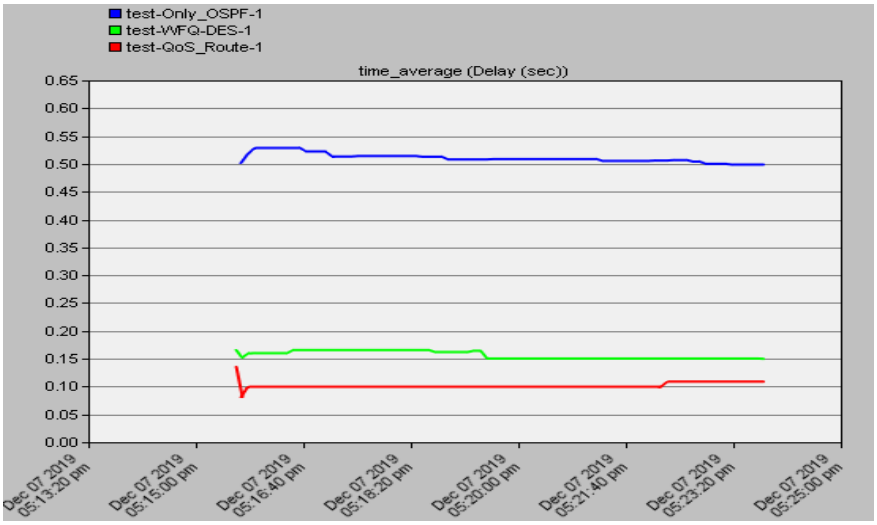

Figure 9. Comparison of average transmission delay in network

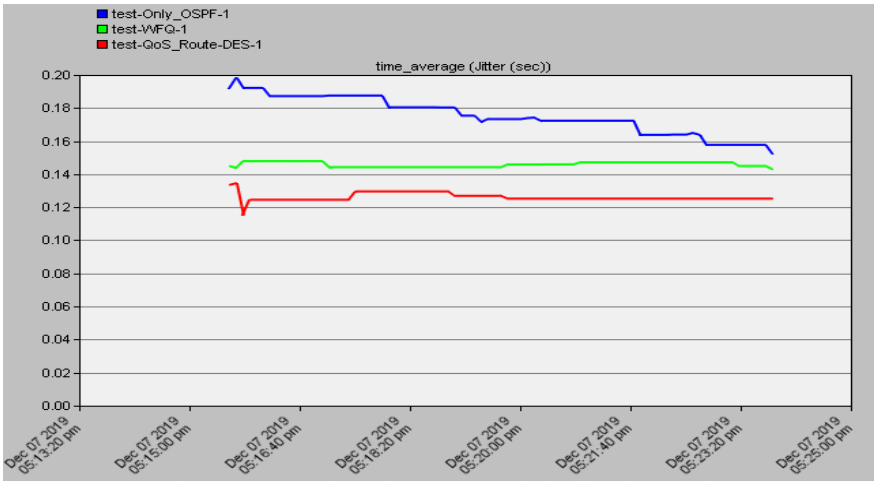

Figure 10. Comparison of average transmission jitter in network

Table 4. A statistical chart for average time delay and jitter of network at different scenarios

	Only_OSPF	WFQ	QoS_Route
Average delay	0.50s	0.15s	0.10s
Average jitter	0.16~0.20s	0.14~0.15s	0.12~0.13s

5. Conclusion

A variety of communication modes and network architectures have arisen in the wake of fast-growing communication-information systems. In this context, heterogeneous networks have become a research focus in communication worldwide. However, the QoS routing studies related to heterogeneous networks are complicated by the heterogeneity and variability of the system. Thus, a network-simulation model has been established in the OPNET Modeler which is based on the composition of the proposed marine-oriented heterogeneous network. The modeling process consists of four major steps: (1) Implement a network layer model, (2) configure an OSPF protocol, (3) design and configure service types, and (4) simulate a QoS routing solution. The experimental results suggest that, by integrating the WFQ algorithm with an extended OSPF protocol, our proposed QoS routing solution can implement effective network resource control and routing optimization.

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